# Using LiDAR to quantify, map, and conserve late-successional and old-growth forest in Maine, USA: Comment

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*Comment on Hagan, J.M., B. Shamgochian, M.M.L. Taylor & J.M. Reed. 2026. Using LiDAR to quantify, map, and conserve late-successional and old-growth forest in Maine, USA. Ecosphere 17(6):e70670. https://doi.org/10.1002/ecs2.70670.*

## 1. Introduction

Mapping late-successional and old-growth (LSOG) forest has become a national undertaking. A 2022 federal executive order directed the first government-wide inventory of mature and old-growth forest (U.S. Executive Order 14072 2022; USDA Forest Service and USDI Bureau of Land Management 2023), and in the same period continental old-growth probability layers (Bruening et al. 2026), structure-based national maps (DellaSala et al. 2022), region-standardized field criteria (Pelz et al. 2023), and global canopy-height products (Potapov et al. 2021) were all produced. A recurring lesson runs through that body of work: because older forest is rare, structurally heterogeneous, and operationalized differently by each program, independent and individually credible maps disagree substantially on both how much LSOG exists and where it sits, and defining the condition at all is a recognized hard problem (Barnett et al. 2023; Gray et al. 2023). That federal effort has also shown that treating maturity and old growth as a multidimensional structural condition rather than a single threshold leaves the estimate strongly dependent on the criteria chosen: the Forest Inventory Growth Stage System builds maturity from a suite of structural indicators across roughly 80 regional vegetation types and places about 45% of federal forest in the mature class and 18% in old growth (Woodall et al. 2023; USDA Forest Service and USDI Bureau of Land Management 2023), an estimate that shifts with the structural thresholds chosen, the national-scale analogue of the definitional sensitivity we quantify here. That disagreement, more than any single map's internal error, is what makes a map's fitness for a particular decision the question that matters. The maps also carry consequences once they are read literally, because a wall-to-wall surface is easily taken as a direct guide to where conservation money should go, stripped of the caveats and uncertainties under which it was built. Maine offers an unusually concrete instance of that general problem, and we treat it here not as a special case but as a worked example of what a cross-map accuracy assessment and an explicit uncertainty statement add before a map anchors spending.

Hagan et al. (2026) provide a valuable, transparent, and timely contribution: a field-trained, wall-to-wall classification of late-successional and old-growth (LSOG) forest across approximately 4.2 million hectares of Maine's unorganized townships, derived from publicly available airborne LiDAR. The authors archived their training data and code, the field effort is substantial, and the resulting map has focused overdue attention on older forest in a working landscape. We were able to reproduce their random forest classifier exactly from the archived data, obtaining an out-of-bag accuracy of 94.2% for the Not-LSOG versus LSOG distinction against their reported 94.1%, with the same most-important predictor (canopy cover above 15 m). Nothing in this Comment questions the competence or the openness of the original analysis, both of which we commend.

Our concern is narrower and methodological, and it follows from how the map is now being used. The classification has become the spatial basis for conservation prioritization at scale: Thompson et al. (2026) build directly on it and estimate that protecting the highest-priority half of the mapped LSOG patches would cost on the order of US $200–300 million, in support of Maine's LD 1529. When a single map becomes the substrate for parcel-level acquisition and emerging carbon and conservation markets, the properties that govern its fitness for that purpose are no longer only its training accuracy. They are (i) its accuracy and agreement relative to independent maps and to ground-based inventory of the same target; (ii) the uncertainty of every quantity derived from it; and (iii) the validity of the trend used to motivate action. A map can be both wrong, in the unavoidable sense that LSOG is a continuous and partly subjective condition forced into discrete classes, and useful for some decisions but not others (Box 1976). Our purpose is to show, using analyses built largely on the authors' own archived data and on independent public data over the same area, what a cross-map accuracy assessment and an explicit uncertainty statement add, and to suggest that they should accompany the map before it anchors large expenditures.

A map built to guide conservation spending must answer three questions: what late-successional and old-growth forest is, how much of it exists, and where it is. The published analysis answers the first only as a canopy-structure proxy and does not yet robustly answer the second or third, and we make four specific points. First, training accuracy is high for every candidate approach and is therefore not the issue (Section 2); what happens beyond the training set is. Second, independent and equally defensible operationalizations of LSOG disagree markedly on where it is, at the very scale at which hectares are prioritized (Section 3). Third, the canopy signal the method relies on is sensitive to tall, big-tree forest but largely blind to the dead wood and disturbance history that distinguish genuine old-growth, so it answers what by mapping a structural proxy rather than old-growth itself, and that proxy is roughly three times more extensive than ground-defined old forest (Section 4). Fourth, how much is reported as a single number without the sampling interval that the ground-based inventory it approximates readily supplies, and that inventory shows the stock stable to rising rather than rapidly declining (Section 5). Section 6 discusses implications.

We write as forest biometricians with a professional interest in how inventory data are used in policy; the analyses below use the authors' archived training data and independent public data over the same area, and all code and derived products are openly archived (Data and Code Availability).

## 2. Training accuracy is high for every approach, and is not the issue

The 463 known-class training hectares carry both the eight LiDAR canopy metrics and a parallel set of ground structural measurements (live and dead basal area, basal area and number of trees ≥ 40 cm dbh, quadratic mean diameter, coarse woody debris, and diameter variability). Using these, we estimated cross-validated AUC (repeated stratified five-fold cross-validation, 40 repeats; intervals from the repeat distribution) for three predictor sets against three binary targets (Table 1). Hagan's eight LiDAR metrics discriminate the field classes very well in-sample: AUC 0.990 for any-LSOG, 0.988 for LS+OG, and 0.966 for old-growth. A canopy-height-only set and a ground-structure set also perform well for the broad classes. The training data are separable and the published classifier is internally sound; we emphasize this because it is the premise that makes the rest of the argument matter. Two features of Table 1 are nonetheless relevant. For old-growth specifically, a canopy-height-only model, which approximates the information a spaceborne canopy-height product carries, is the weakest of the three (AUC 0.910 [0.868, 0.933]), while ground structure is the strongest. And the full LiDAR model's high ranking ability coexists with the operating-point performance the authors report: in their own out-of-bag confusion matrix, only 29.4% of true old-growth hectares were classified as old-growth-like. The authors are themselves candid about this, labelling the modeled class old-growth-like precisely because those hectares were predicted, not field-verified, as old-growth, and noting that distinguishing old-growth from late-successional forest is where their canopy metrics are weakest; our point is that this conceded weakness is consequential, because it is the old-growth class that the prioritization treats as the highest-value target. High AUC and low operating accuracy are reconcilable under class imbalance and overlap, but for a map used to select specific hectares, it is operating performance on the rare, high-value class that governs the decision. The point is sharpened by the paper's field validation, which contained no true old-growth sites at all (Hagan et al. 2026, Table 6): the rarest and highest-value class, the one a conservation program would most want to locate, was evaluated only against the authors' own training labels and never field-verified.

This rare-class shortfall reflects the algorithm's handling of class imbalance more than the data. Refitting the published random forest on the archived training data with standard class balancing raises old-growth operating accuracy from 23.5% (our unbalanced reproduction, consistent with the authors' reported old-growth detection) to 70.6% under balanced sub-sampling, and to about 82% under a lowered voting threshold, at a negligible cost to overall accuracy (0.87 to 0.84), more than doubling the mapped old-growth area from about 1.0% to between 1.9% and 2.3% of forested hectares (on the order of 109,000 to 192,000 acres under balancing), with any-LSOG rising from 21.9% to 32.3% on the comparison grid (Supporting Information). The wall-to-wall area a single classifier reports is therefore sensitive to a routine modeling choice, a further reason it should not by itself anchor parcel-level expenditure. We do not claim the balanced classifier is the more correct one: class balancing trades omission of the rare class for commission, and which way the truth lies cannot be settled from the training labels alone. That is the point. The mapped area swings with a routine modeling choice that only independent field validation of the old-growth class could adjudicate.

A deeper issue underlies the rare-class shortfall: the training data are not a probability sample of the landscape. The 463 known-class training hectares were selected purposively, by expert judgment and a search for features associated with older forest, and anchored on named old-growth sites such as Big Reed Reserve, the authors' stated primary source for old-growth training data (Hagan et al. 2026). The resulting set is enriched in older forest roughly twofold for the broad LSOG class (about 39% of training hectares versus 20% of the mapped landscape) and roughly fivefold for the rare late-successional and old-growth classes (about 21% versus 3.9%), and the random forest was fit with default settings without class weighting. The objection is not that a supervised classifier learns from known examples, which is unavoidable and appropriate; it is that the training prevalence does not match the landscape and was not corrected for. A classifier trained on a class distribution that does not match the landscape, and not corrected for it, has an operating point and an implied wall-to-wall prevalence that are not anchored to the landscape's true class frequencies; this is both why the rare-class operating accuracy is low and why the mapped area shifts substantially under routine rebalancing. The remedy is a balanced, representative training set that draws both LSOG and non-LSOG observations from the landscape, or a design-based probability sample such as FIA that is representative by construction and that we use here as the area anchor. Independent, systematically measured field networks that contain true old-growth were, moreover, already available for exactly this purpose: the Maine Natural Areas Program and The Nature Conservancy ecological reserve network, monitored on fixed plots across 36 reserves, and the long-running Baxter State Park Scientific Forest Management Area continuous forest inventory. These standing, independent datasets, rather than a purposive field campaign that included no true old-growth validation sites, are the appropriate basis for verifying the map, including the rare class the authors' own validation could not test.

**3. High training accuracy does not produce map agreement**

The decisive test for policy use is not how well a model separates its own training plots but whether independent, equally defensible operationalizations of LSOG agree on the landscape. We compared three remote-sensing classifications that differ in sensor, training data, and target definition, over the study area on a common 100 m grid: the reproduced Hagan airborne-LiDAR classifier; an FIA-field-structure class predicted from Potapov Global Ecosystem Dynamics Investigation (GEDI)-calibrated spaceborne canopy height; and the Oak Ridge National Laboratory (ORNL) national mature-and-old-growth product (Bruening et al. 2026; old-growth stratum). The two structure-resolution maps differ 1.6-fold in extent (Hagan any-LSOG 21.9% versus canopy height 14.0%, each as a fraction of forested hectares on the common comparison grid; the published map reports its three LSOG classes as 15.8% Transitioning LS, 3.0% LS, and 0.9% old-growth-like, totaling 19.7% of the full study area, Hagan et al. 2026 Table 7) and overlap only modestly (Cohen's kappa 0.22). The ORNL old-growth stratum is a coarser, continental product that flags a much larger area (36.1%, and 65% for its mature-plus-old-growth stratum); including it widens the range to 2.6-fold (Fig. 1a-c). Spatially the three overlap poorly: only 2.7% of the hectares flagged by any map are flagged by all three, and 73% are flagged by a single map (Fig. 1d). The ORNL stratum shares little spatial information with either structure map (kappa near zero); rather than evidence that any one map is wrong, this is a cross-scale caution that a continental old-growth product and a state airborne-LiDAR product, each credible in its own domain, identify largely different ground. We separately reproduced the privately held

Seven Islands classification and found it effectively identical to the published model (pixel-level kappa 0.97 across approximately 290,000 ha), which confirms that the disagreement among methods is genuine rather than an artifact of our reproduction.

None of the three products is ground truth, including ours; all are remote-sensing proxies for the ground-measured structure that defines LSOG, and each embeds different definitional and sensor choices. The point of the comparison is not that the Hagan map is wrong and another is right. It is that several credible, high-training-accuracy operationalizations of the same concept place markedly different amounts of forest in markedly different places, and that spread is itself the quantity a high-stakes user needs to see. Accuracy issues of this kind are well documented even among global canopy-height products built from the same sensors (Moudrý et al. 2024).

The consequence for prioritization is direct. If a program protects the highest-priority hectares, the overlap of the protected sets selected by the Hagan map versus the canopy-height map is only 0.16 to 0.30 (Jaccard) for the top 5% to 20% of hectares; that is, 70% to 84% of the prioritized ground differs depending on which equally defensible map is used (Fig. 1). Wherever a map of this kind steers acquisition, the choice of map, not only the choice of parcels, would determine much of what is bought: a large program guided by one credible map would protect substantially different forest than the same program guided by another. In Maine, where a prioritization on the order of $200–300 million is contemplated, that sensitivity is not abstract, but the implication is general and bears on any policy or acquisition strategy built on a single LSOG surface. Each map can have excellent training accuracy and the maps can still disagree on most of the ground, because the disagreement lives in the definitional and methodological choices rather than in the fit to any one training set.

## 4. The canopy signal maps big-tree forest, not old-growth

Old growth is defined ecologically by large old trees, structural complexity, and abundant dead wood. Using the archived plots, we asked how well the eight LiDAR canopy metrics predict these defining attributes (cross-validated $R^2$). They predict large-tree basal area moderately ($R^2$ = 0.68) but are largely blind to dead wood: coarse woody debris volume $R^2$ = 0.20 and standing dead basal area $R^2$ = 0.24 (Table 3). The blind spot is not confined to the canopy metrics; it reaches into the field protocol that trained the map. In the random-forest classifier of the authors' own RAP v2.0 rapid-assessment protocol (Shamgochian et al. 2025), large standing dead trees rank last of seventeen metrics in importance and large downed logs thirteenth, even though that document calls large logs one of the best indicators of old-growth, while harvest-history evidence (sawn stumps, skid trails) is among the strongest predictors. The dead-wood component that distinguishes old-growth is therefore under-weighted by the training labels as well as by the LiDAR. A classification keyed on canopy height and cover therefore registers tall, continuous canopy whether it is produced by an old, structurally complex stand or by an 80- to 100-year-old fast-growing stand, a failure mode the authors note for *Populus*. The scale of the resulting ambiguity is quantifiable from ground inventory (Section 5): roughly 12.5% of Maine forest carries substantial large-tree basal area, but only about 3.9% meets a stand-age criterion for old forest, so big-tree forest is about three times more extensive than ground-defined old forest. The large-tree threshold is illustrative rather than definitional, and the two criteria measure different things; we use the contrast only to bracket the magnitude of the over-inclusion a height- or canopy-based classifier produces relative to a structural or age-based definition, which is consistent with the among-method spread in Section 3. The over-inclusion is compounded by treating LSOG as a single, universal target. Old-growth structure differs by forest type, so a threshold tuned to tall, large-diameter northern hardwood or white pine systematically misses the low-stature spruce-fir, northern white-cedar, and peatland old-growth that never attains that canopy signature. In the ecological reserves, a single type-agnostic large-tree threshold flags 32% of plots, whereas externally published forest-type-specific structural criteria (Pelz et al. 2023) flag 67%, with spruce-fir rising from 29% to 68%; Maine's own documented old-growth inventory is itself organized by seven forest types, each with a distinct age and size signature (Maine State Planning Office 1983). A defensible classification therefore needs ecoregion- and forest-type-specific criteria rather than one universal definition, a point the canopy metrics, tuned to a single signal, cannot accommodate. A 1 ha mapping unit also limits the other end of the size scale: because Maine's genuine old-growth persists largely as small remnants, often only a few hectares, a 1 ha classification cannot resolve sub-hectare patches at all, so the same grid that over-includes big-tree forest under-resolves the smallest true old-growth. Mapping those remnants is a task for high-resolution airborne LiDAR validated at true field coordinates, not for a coarser wall-to-wall product. The choice of grain is not neutral for the headline number, either. The published map averaged its 1 m^2 canopy metrics to the hectare and classified once at that single scale, without testing sensitivity to resolution; the finer structure that distinguishes a small old patch is therefore averaged into its matrix before any class is assigned. Aggregating a 10 m LSOG probability surface to coarser cells lowers the flagged LSOG share monotonically, from about 22% at 10 m to 17.5% at the hectare and 14.9% at 200 m; this is a consequence of averaging the probability surface and re-applying a fixed threshold rather than a property of older forest itself, and we draw from it only the operational point that a single percentage is specific to the grain and thresholding rule at which it was produced, as area-based LiDAR estimates are known to be in these northeastern multicohort forests (Hayashi et al. 2016). Any comparison among maps, or between a map and the inventory, must therefore hold resolution fixed.

## 5. Ground-based inventory: the estimate, with the interval the map omits

Airborne LiDAR and spaceborne canopy height are proxies for the ground-based structural measurements that define LSOG; TreeMap (Riley et al. 2021) imputes those same measurements from the FIA plots themselves, so it is a second FIA-anchored accounting rather than an independent map. The USDA Forest Inventory and Analysis (FIA) program provides the measurements under a probability sample, which permits design-based estimates with sampling-error variance. Using rFIA design-based estimation (post-stratified, with FIA estimation-unit areas; Stanke et al. 2020) over all Maine inventory years, the area of older forest by transparent ground criteria is (2024, 95% CI): stand age ≥ 100 yr, 12.1% [10.9, 13.2] of forestland; stand age ≥ 120 yr, 3.9% [3.3, 4.6]; stand age ≥ 150 yr, 0.7% [0.4, 1.0]; and live basal area in trees ≥ 40 cm dbh exceeding 30 $ft^2$/ac, 12.5% [11.4, 13.7] (Table 4). In the northern timberland units that approximate the study area, the stand-age ≥ 120 yr estimate is 4.2% [3.3, 5.1]. The point estimate for older forest brackets the published LS+OGL figure of 3.9% (161,881 ha), which is reassuring, but the original analysis reported that figure without an interval; the design-based estimate places it in a range of roughly 3.3 to 4.6%. Carrying that interval forward is not a statistical nicety; it is among the most decision-relevant pieces of information a high-stakes conservation program can have, because the width of the interval, not the point estimate alone, tells a program how much weight the underlying number can bear before money moves. Agreement on the amount, moreover, is not agreement on location: the inventory and the map can report nearly the same regional total while disagreeing on which specific hectares are old, and it is the specific hectares, not the total, that a parcel-level acquisition program buys. We make this concrete by regenerating the published classification from its own archived inputs, the same random forest applied to the public hectare-level LiDAR canopy statistics for all 4.28

million hectares, which reproduces the reported extent to within about two percentage points (22.0% any-LSOG against 19.7%, the residual reflecting default classifier settings and grid alignment rather than the authors' exact configuration), and then thresholding our FIA-anchored 10 m embedding map to flag the identical landscape fraction so that any disagreement is about location alone. At matched amount the reproduced classifier and the embedding map agree on only 7.7% of the landscape, each flagging roughly 14% the other does not (Cohen's kappa 0.19). Neither is ground truth and neither is being held up as correct; the comparison isolates the locational uncertainty that persists when two independent methods agree on the amount, and it is a lower bound. That the two perform comparably at independently located old-growth (Section 4) indicates this reflects genuine ambiguity in the working forest rather than one map being broadly wrong. The amount can coincide while the location does not. Because FIA stand age is a modeled field attribute, poorly defined in the uneven-aged stands that dominate this region, we treat the structural criterion, live basal area in large trees, as the primary ground measure and the age thresholds as corroborating rather than definitive; the structural and age estimates bracket a consistent range, and the policy-relevant conclusions below rest on the structural measure, which carries no assumption about a stand's assigned age.

Applying the same multi-axis LSOG definition to the FIA probability sample with design-based estimation gives the disputed quantity directly: under the integrated structural proxy, late-successional-and-old-growth forest covers 14.1% [12.9, 15.3] of Maine forestland (2019–2023 FIA panel, 3,125 plots), and requiring all four structural axes, each on a transparent and deliberately conservative criterion, reduces true LSOG to 3.1% [2.5, 3.7] (Table 4): live large-tree structure (basal area ≥ 30 $ft^2$/ac in trees ≥ 40 cm, about 16 in, dbh), dead wood (standing-dead basal area ≥ 5 $ft^2$/ac), compositional maturity (at least half of live basal area in long-lived late-successional species), and temporal continuity verified from the full Landsat disturbance record. These are lower benchmarks calibrated to regional reference stands rather than maxima, so the qualifying share is a floor on what a stricter definition of large-tree or dead-wood structure would return. The reproduced LiDAR map's 21.9% any-LSOG is the highest of the operational maps and exceeds the design-based any-LSOG estimate of 14.1%. The spaceborne canopy-height map (14.0%) coincides almost exactly with that design-based estimate, so two routes that differ in sensor and method converge near 14% while the airborne map sits well above them; the TreeMap imputation (7.8%) sits below, and the strict four-axis estimate (3.1%) far below. The appropriate reporting unit remains the design-based estimate with its interval, against which the airborne map is the highest of the operational estimates. This ordering reflects a principle rather than a coincidence of these particular numbers. For the amount of LSOG, a design-based estimate from a probability sample is the reference quantity, with a known and quantifiable sampling error, that any wall-to-wall map of the same attribute is obliged to reproduce. When a map departs from that estimate, the discrepancy is a property of the map and not of the inventory, and the burden falls on the map to reconcile with the design-based total rather than the reverse. A map's distinctive contribution is spatial, telling us where that total is most likely to fall; it does not overturn the total itself. The contrary objection, that FIA lacks the resolution to see LSOG and that a wall-to-wall map is therefore required, conflates spatial resolution with statistical resolution of a total. FIA cannot say which hectare is old, but that is the map's task; for how much old forest a region holds, the probability sample is the design-unbiased reference, and rarity widens its interval without biasing its estimate. The federal analysis ran short of plots only because a national stratification spreads them thin; concentrated on Maine, the same sample supplies several thousand plots and bounds even the oldest class (0.7% [0.4, 1.0]). The map does not escape the rarity it is invoked to solve, it conceals it, reporting a rare-class area that is both biased by a non-representative training sample and given without an interval. Tellingly, the published LS+OGL total of 3.9% falls inside the design-based interval [3.3, 4.6]: the inventory resolves the amount, and on that amount the two agree. What remains in dispute is the broader any-LSOG total and the location of LSOG, and a finer-grained but design-biased map is not the instrument that settles either.

The design-based series also addresses the loss premise directly. The map is motivated in part by rapid LSOG loss, reported at 1.37% per year overall and 2.19% per year on commercial timberland. That figure is a gross harvest flux: it counts mapped LSOG hectares that subsequently experienced canopy removal in the Global Forest Watch record, and it does not net out ingrowth, the younger stands that mature into older condition each year. The FIA design-based stock moves the other way. Over 2003–2024, older forest increased across the operational age and structure measures, with confidence intervals excluding zero: stand age ≥ 100 yr at +0.19% of forestland per year [0.15, 0.23], stand age ≥ 120 yr at +0.065% per year [0.054, 0.075], and large-tree basal area at +0.17% per year [0.15, 0.18], while total forestland area was essentially flat (Fig. 2). The net-stock increase holds on the large-tree basal-area measure alone, which makes no use of FIA's age assignment, so the result does not depend on stand-age modeling. The northern units show the same increases; only the oldest class (age ≥ 150 yr) in the north declines slightly (-0.019% per year [-0.031, -0.008]). Both statements can be true at once: LSOG stands are harvested at roughly 2% per year, and the total older-forest stock is not declining, because aging more than replaces the hectares removed. Where a rate of loss is invoked to convey urgency, the flux and the stock should be distinguished. The authors themselves note they had no way to estimate the rate at which forest grows into the LSOG condition, so the reported loss rates are gross removals from a canopy-change product without the offsetting recruitment that the design-based stock captures directly.

We note finally that the publicly available spaceborne canopy-height products covering the study area (Potapov et al. 2021; Lang et al. 2023), although cited in passing in the original introduction, are not used as an independent benchmark. They are coarser than airborne LiDAR but global, repeatable, and free, and they provide the kind of independent cross-check, illustrated in Section 3, that a single airborne map otherwise lacks.

## 6. Implications

None of the above argues against mapping LSOG with LiDAR, nor against conserving older forest in Maine, which we support. It argues that a single classification, however well trained, is not a sufficient basis for parcel-level expenditures of the magnitude now contemplated, and that three practices should accompany its policy use. First, LSOG extent should be reported as a range across methods and with the design-based sampling interval, rather than as a single number; the ground-based estimate of older forest, 3.9% [3.3, 4.6], is an example. The qualifying criteria themselves should be ecoregion- and forest-type-specific rather than universal, because a single structural threshold misclassifies the low-stature old-growth that defines much of Maine's spruce-fir and cedar. Second, before hectares are prioritized for acquisition or enrolled in markets, the mapped total should be reconciled with the design-based inventory estimate of the same quantity, the map checked against at least one independent product, and candidate parcels field-verified, ideally against the standing independent field networks that already contain true old-growth, the ecological reserves, the Baxter inventory, and the fifteen confirmed natural old-growth stands documented on State lands, 1,553 acres concentrated in Baxter and the northern Deboullie unit (Maine State Planning Office 1986), and at the parcel scale, as the authors themselves recommend for management decisions. This matters because the downstream prioritization already delineates tens of thousands of patches from the single classification and ranks them for acquisition without propagating the map's uncertainty (Thompson et al. 2026), so a probabilistic surface with explicit bounds, rather than one binary map, is the appropriate substrate for parcel-level spending. Because most older forest across the region sits on private land, this same uncertainty bears on working-lands easements, payments for ecosystem services, and conservation planning on those lands, not on fee acquisition alone; the dominance of private ownership is itself why accurate, uncertainty-aware maps are needed to plan conservation creatively at a regional scale rather than within a single state. Third, where a rate of loss is invoked, the harvest flux and the net stock should be distinguished, because they point in opposite directions here. The constructive next step, which we develop in a companion analysis, is a repeatable, FIA-anchored, multi-axis classification of LSOG with an explicit cross-map accuracy assessment; that work is beyond the scope of a Comment but follows directly from it. A management-facing synthesis of these limitations, paired with recommendations and our design-based findings, is tabulated in Weiskittel (2026). The Hagan et al. map is a useful and reproducible first effort and a real contribution to attention on older forest. It is not, as it stands, a sufficient basis for the parcel-level acquisition decisions now being built on it: trained on a purposively assembled, LSOG-enriched sample rather than a probability sample, fit with a default classifier that detects the rare old-growth class fewer than 30% of the time in the authors' own accounting, never field-validated for that class, and reported as a single area with no interval and no agreement with independent maps, it does not yet robustly answer how much LSOG exists or where it is. The cross-map accuracy assessment, the design-based interval, and the uncertainty layer are therefore not refinements to add later; they are prerequisites before a map of this kind directs hundreds of millions of dollars.

**Competing Interests**

The authors are developing an independent, FIA-anchored multi-axis LSOG classification (the companion analysis referenced here), which could be regarded as an alternative to the product discussed. This Comment is offered as a methodological critique in that context, and all supporting analyses use the original authors' archived data and public data over the same area. The alternative we are developing is held to the same standard we apply here: it equally requires independent field validation of the old-growth class, and our own analyses indicate that even modern wall-to-wall products fall short of replacing design-based estimation for this target: an open Sentinel-based satellite-embedding model reaches a cross-validated area-under-curve near 0.82, and fusing multiple modern sources lifts it to about 0.87, both well above a single coarse canopy-height layer (about 0.67 to 0.70) yet still imperfect for a class this rare; these benchmarks are documented in the companion technical report (Weiskittel 2026). The argument is therefore not that one map outperforms another, but that no remote-sensing map, ours included, can substitute for design-based estimation of how much LSOG exists.

**Data and Code Availability**

All analyses, code, and derived products supporting this Comment are archived at Zenodo (concept DOI 10.5281/zenodo.20614496, resolving to the latest version). Hagan et al.'s training data and code are at Zenodo (10.5281/zenodo.19696494). FIA data are from the USDA FIA DataMart (apps.fs.usda.gov/fia/datamart), accessed June 2026.

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## Tables

**Table 1. Cross-validated AUC (mean [95% interval]) recovering the field-assigned class from three predictor sets, on the 463 known-class training hectares.**

| Target (prevalence) | Hagan 8 LiDAR | Canopy height only | Ground structure |
|---|---|---|---|
| any-LSOG (0.39) | 0.990 [0.988, 0.992] | 0.984 [0.981, 0.987] | 0.966 [0.962, 0.969] |
| LS + OG (0.21) | 0.988 [0.984, 0.990] | 0.977 [0.973, 0.982] | 0.970 [0.967, 0.973] |
| old-growth (0.04) | 0.966 [0.954, 0.977] | 0.910 [0.868, 0.933] | 0.974 [0.966, 0.981] |

**Table 2. Wall-to-wall any-LSOG area and pairwise spatial agreement across three independent remote-sensing maps over the study area. TreeMap is an FIA imputation and is reported with the design-based estimates (Table 4), not here.**

| Method | any-LSOG (% of area) | kappa vs Hagan | kappa vs canopy |
|---|---|---|---|
| Hagan (airborne LiDAR canopy) | 21.9 | -- | 0.22 |
| Canopy height (Potapov/GEDI) | 14.0 | 0.22 | -- |
| ORNL old-growth stratum (Bruening) | 36.1 | -0.02 | -0.07 |
| 3-way: % of flagged area agreed by all three | 2.7 | | |

**Table 3. Cross-validated $R^2$ for predicting ground structural attributes from the eight LiDAR canopy metrics.**

| Structural attribute | CV $R^2$ from LiDAR |
|---|---|
| Live basal area in trees ≥ 40 cm dbh | 0.68 |
| Number of trees ≥ 40 cm dbh | 0.59 |
| Diameter variability (CV of dbh) | 0.51 |
| Standing dead basal area (snags) | 0.24 |
| Coarse woody debris volume | 0.20 |

**Table 4. FIA design-based LSOG and older-forest area with 95% CI. Single-axis age and structure criteria are 2024 with statewide, northern-unit, and 2003–2024 trend estimates; the multi-axis LSOG classes are the 2019–2023 panel (statewide). TreeMap, an independent FIA imputation, gives any-LSOG 7.8%.**

| Domain (ground criterion) | Statewide % [95% | Northern units % | Trend (%/yr, |
|---|---|---|---|

| | CI] | [95% CI] | statewide) [95% CI] |
|---|---|---|---|
| Integrated any-LSOG (structural proxy) | 14.1 [12.9, 15.3] | -- | -- |
| True LSOG, all four axes (incl. continuity) | 3.1 [2.5, 3.7] | -- | -- |
| Stand age ≥ 100 yr | 12.1 [10.9, 13.2] | 12.2 [10.7, 13.7] | +0.19 [0.15, 0.23] |
| Stand age ≥ 120 yr | 3.9 [3.3, 4.6] | 4.2 [3.3, 5.1] | +0.065 [0.054, 0.075] |
| Stand age ≥ 150 yr | 0.7 [0.4, 1.0] | 0.6 [0.2, 1.0] | +0.000 [-0.005, 0.006] |
| Large-tree BA ≥ 30 $ft^2$/ac (≥16 in) | 12.5 [11.4, 13.7] | 9.8 [8.4, 11.2] | +0.17 [0.15, 0.18] |

**Figure Legends**

Fig. 1. Three independent remote-sensing operationalizations of LSOG over Maine on a common 100 m grid disagree in amount and location: (a) Hagan airborne LiDAR, any-LSOG 21.9%; (b) Potapov/GEDI spaceborne canopy height, 14.0%; (c) ORNL old-growth stratum (Bruening et al. 2026), 36.1%; (d) number of maps agreeing on LSOG per cell (0–3). Only 2.7% of the hectares flagged by any map are flagged by all three, and 73% are flagged by a single map. The two structure-based maps agree only modestly (Cohen's kappa 0.22) and the ORNL old-growth stratum is uncorrelated with either (kappa near zero). Each panel is clipped to the Maine land area.

Fig. 2. FIA design-based older-forest area (stand age ≥ 120 yr) for Maine, 2003–2024, with 95% confidence intervals; the stock rises while total forestland area is flat.

**Figure 1**

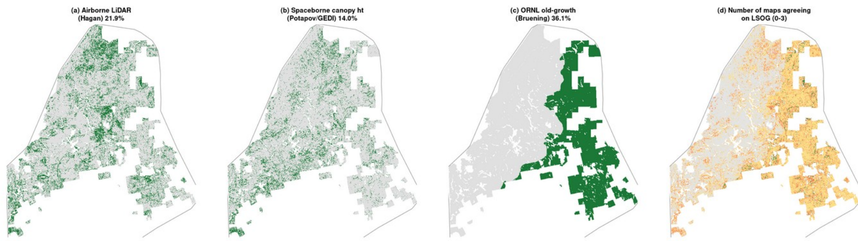
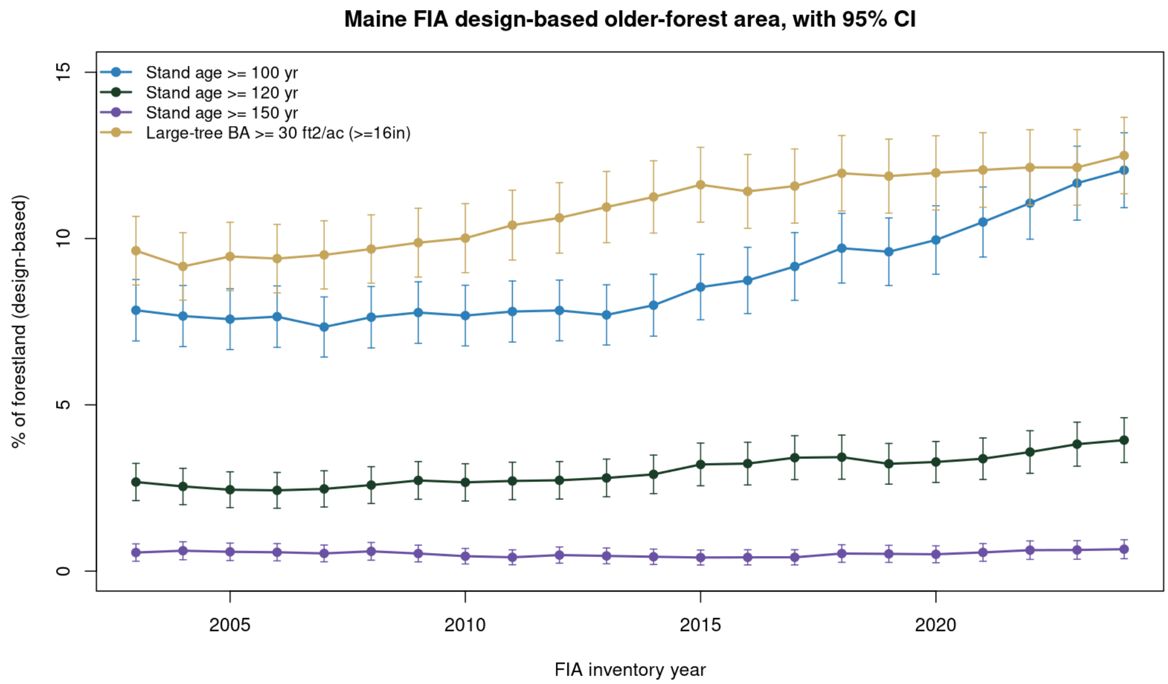

(a) Airborne LiDAR
(Hagan) 21.9%
(b) Spaceborne canopy ht
(Potapov/GEDI) 14.0%
(c) ORNL old-growth
(Bruening) 36.1%
(d) Number of maps agreeing
on LSOG (0-3)

**Figure 2**


Maine FIA design-based older-forest area, with 95% CI
Stand age >= 100 yr
Stand age >= 120 yr
Stand age >= 150 yr
Large-tree BA >= 30 ft2/ac (>=16in)
% of forestland (design-based)
0
5
10
15
2005
2010
2015
2020
FIA inventory year